EHLERS-DANLOS SYNDROME

http://www.aocd.org

Ehlers-Danlos Syndrome (EDS) is a group of inherited connective tissue disorders that are caused by mutations in genes encoding the structure or processing of collagen. Collagen is a vital protein involved in connective tissue support, strength and elasticity of skin. It is an important building block that provides structure to many organs and tissues in our body. Mutations in collagen, as seen in EDS, cause widespread complications involving skin, muscles, bones and blood vessels. The classic symptoms of EDS are skin hyperextensibility, joint hypermobility, easy bruising and skin fragility.

The 2017 International Classification for EDS identifies 13 different subtypes based on genetics and clinical manifestations. The most common subtype is hypermobile EDS, followed by classical EDS and vascular EDS. The other, more rare subtypes, include classical-like EDS, cardiac-valvular EDS, arthrochalasia EDS, dermatosparaxis EDS, kyphoscoliotic EDS, brittle cornea syndrome, spondylodysplastic EDS, musculocontractural EDS, myopathic EDS and periodontal EDS. Diagnosis of EDS is made through detailed history and physical exam. In order to identify the specific subtype of EDS, tissue **biopsy** and genetic studies are performed.

The skin findings and symptoms vary depending on the subtype and severity of EDS. A common feature is skin hyperextensibility and fragility, meaning the skin is loose, stretchy and tears easily. The skin is also predisposed to complications with wound healing leading to excessive bruising and "cigarette-paper," thin, wide atrophic scars. People with EDS may also develop other cutaneous symptoms such as molluscoid pseudotumors, subcutaneous spheroids, petechiae and translucent skin.

EDS is a systemic disease and affects multiple organs, in addition to the skin. Several musculoskeletal symptoms occur such as joint hypermobility, instability and dislocation, scoliosis and muscle pain. Other systemic symptoms include, but are not limited to, chronic fatigue, palpitations, heart valve dysfunction, periodontal disease, gastrointestinal disorders and ocular disorders.

There is no cure for EDS and treatment is supportive to improve quality of life. Treatment options include physical therapy, pain control and potential surgery for damaged joints. Ascorbic acid (Vitamin C) has been shown to reduce bruising and improve wound healing. Orthopedic instruments, such as braces or casting, can help stabilize joints and prevent further damage. People with EDS require a multidisciplinary team approach to ensure routine evaluation of the skin, eyes, heart and joints.